

Eco-Loop Building Agents

A supervisory controller that commissions itself onto a building it has never seen — and refuses to deploy when nothing earns its place.

Problem Statement ID — 1

Problem Statement Title — Eco-Loop Building Agents

Theme — Smart Buildings & Energy Management

PS Category — Software

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FULL YEAR, DOE MEDIUM OFFICE, CHICAGO

−5.33% electricity

−8.66% peak demand

56.0 → 51.5 unmet hours (lower than baseline)

Closed loop on a live EnergyPlus instance · 13-tool MCP server, stdio and HTTP · commissions itself onto a building it has never seen · repairs a model EnergyPlus
refuses · 55 tests

IDEA TITLE

Eco-Loop — a supervisory controller that commissions itself, and refuses when nothing earns its place

Advanced control already saves 15–30%. It reaches under 5% of buildings, because every deployment costs weeks of engineering — the blocker is commissioning, not the algorithm.

WHAT IT DOES

- 1 Discover** — reads the model's own wiring to find what can be actuated
- 2 Survey** — runs the building untouched to see where the energy goes
- 3 Trial** — simulates each candidate measure against that baseline
- 4 Deploy or refuse** — keeps only what actually helps; ships nothing otherwise

Control is written into a live EnergyPlus instance every timestep through its runtime API — not by editing a file and re-running.

WHY IT IS DIFFERENT

A measure must earn its place applicability is not benefit. On one building the obvious measure targeted 16.6% of load, was fully actuatable, and made things 2.14% worse. Only a trial caught it.

It refuses two of three building-and-climate pairs deploy nothing, and say why.

Comfort is scored where the controller cannot reach unmet hours are gameable — a zone held at 29 °C reports zero. A test proves it.

We can prove where the LLM helps a perfect-foresight arm bounds the task, so a null result is visible instead of hidden.

−5.33%

electricity, full year

−8.66%

peak demand

56.0 → 51.5

unmet hours, improved

0

measures shipped where none helped

TECHNICAL APPROACH

OPEN-SOURCE LLM Qwen2.5-3B via Ollama, local

sets one number a day: the supply air ceiling, from a degraded 24 h forecast
schema-constrained JSON · 368 decisions/yr · 0 failed · median 7.6 s



MCP SERVER 13 tools, stdio + streamable HTTP

one capability layer, two surfaces: the agent calls it in-process, external clients over MCP
evaluate_policy lets any client close the loop without touching this codebase

policy down



DETERMINISTIC CORE runs every timestep, no inference

Guardian clamps every policy into a comfort band the controller cannot move
Guideline 36 supply-air reset · commissioning by trial · self-repair of broken models



ENERGYPLUS 26.1 live instance, runtime Python API

2 callbacks per timestep: read state and write actuators in, record telemetry out
52,560 timesteps a year in 14 s · epJSON mutation · paired baseline on one clock



telemetry up

Python 3.11 · EnergyPlus 26.1 runtime API (pyenergyplus) · epJSON model mutation · Qwen2.5-3B on Ollama (local, open-source) · Model Context Protocol (FastMCP) · Pydantic contracts · pytest, 55 tests · matplotlib report

The model is never in the inner loop: 52,560 inferences a year at 7.6 s each is 111 hours against a 14-second simulation. It sets one number a day; the deterministic core runs every timestep and the Guardian bounds it.

FEASIBILITY AND VIABILITY

BUILT AND RUNNING TODAY

- Whole system runs on one laptop — an M1 with 5.3 GiB of VRAM
- Annual simulation with full telemetry: 14 seconds
- 55 tests, including real simulations, green in CI
- One command regenerates every published number in ~3 minutes

RISKS, AND WHAT ANSWERS THEM

EnergyPlus API fails silently → four known traps handled; totals reconciled against E+'s own report by test

Model slow, wrong or offline → never in the inner loop; last-good plan; a test kills it mid-run

A measure harms the building → trial verification, then refusal

Comfort quietly traded away → fixed band the controller cannot move, enforced by the Guardian

THE RESULT WE DID NOT EXPECT, AND KEPT

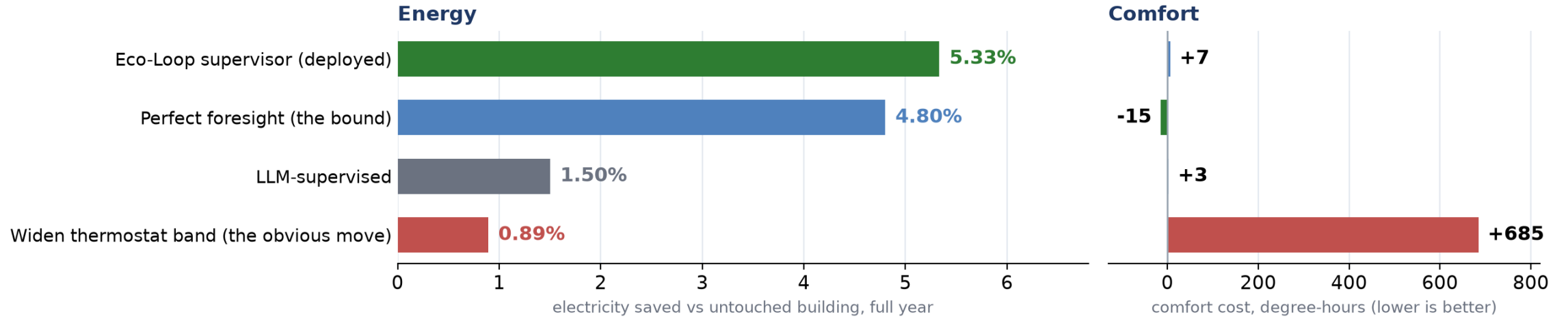
The LLM-supervised arm does not beat the deterministic controller: -1.50% against -5.33% over a year.

Before blaming the model we bounded the task. A perfect-foresight arm — the same controller handed the day's forecast peak, which is exactly the anticipation the model was asked for — lands on the reactive controller to within 0.03 points over a season. There is no headroom on this measure for anything to find.

Against the untouched building alone the agent saves 1.5% and could be reported as a success. It is only visible as a null result because the deterministic bar exists. The model's value here is commissioning and refusal, not energy — and the catalogue of measures, not the framework, is now the limit.

ARTIFACTS

ABLATION LADDER — five arms, identical weather, run period and timestep, compared on one asserted-identical clock



SELF-COMMISSIONING — the same code pointed at a building it was not built for

```
$ ecoloop commission --model RefBldgSmallOfficeNew2004_Chicago.idf
heating 0.0% cooling 7.0% fans 16.6% lights 24.0% equipment 37.1%
[ ] supply_air_reset nothing to actuate; no heating handle in this model
[ ] hvac_availability tried it: +2.14% electricity, not worth it
deploys nothing; no measure in the catalogue earns its place here
```

<https://github.com/AryanAT13/honeywell> · source, 9 arch decision records, the generated evidence report, and the agent's 390-record decision journal so the LLM arm replays exactly without a model server

RESEARCH AND REFERENCES

ASHRAE Guideline 36-2021

High-performance sequences of operation for HVAC systems — the supply-air reset and trim-and-respond logic implemented here

ASHRAE 90.1 Appendix G

Unmet load hours as a compliance limit; our baseline sits at 56 h against the 300 h allowance

DOE Commercial Reference Buildings

Medium and small office prototypes used as the baseline models, unmodified

EnergyPlus 26.1 Python API

nrel.github.io/EnergyPlus — runtime callbacks, actuators, and the epJSON schema used for model mutation

Model Context Protocol

modelcontextprotocol.io — tool specification; served here over stdio and streamable HTTP via FastMCP

Qwen2.5-3B-Instruct · Ollama

open-source model run locally with JSON-schema-constrained decoding; ISHRAE 2014 New Delhi weather from climate.onebuilding.org

VERIFY IT YOURSELF

make setup · make evidence — regenerates every published number in about three minutes, with no model server needed · make test — 55 tests

Method note — forecasts given to the model are deliberately degraded with error growing by lead time, because the weather file is perfect foresight. Delhi results are a climate-sensitivity probe, not a claim about Indian buildings: the US prototype is undersized for that climate.